

Deeper Look at ML

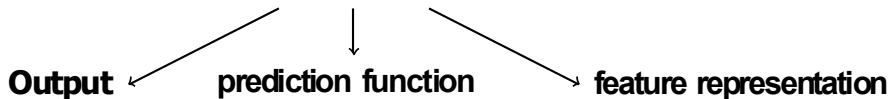
- ▶ Apply a prediction function to a feature representation of the “sample” to get the desired output:

$$f(\text{apple image}) = \text{“apple”}$$

$$f(\text{tomato image}) = \text{“tomato”}$$

$$f(\text{cow image}) = \text{“cow”}$$

$$y = f(x)$$



Training: given a training set, estimate the prediction function f by minimizing the prediction error

Testing: apply f to never before seen test example x and output predicted value $y = f(x)$

The Underlying Abstraction

$$y = f(x)$$

- ▶ What are x, y for spam detection?
- ▶ What are x, y for image classification?
- ▶ What are x, y for sentiment analysis?
- ▶ What are x, y for < insert your problem here > ?

The Underlying Abstraction

$$y = f(x)$$

- ▶ What are f , x , y for the classification tasks?
- ▶ x is often a vector (column matrix).
- ▶ y is either 0 or 1 (binary) or $\{1, 2, \dots, p\}$ (multiclass)

The Underlying Abstraction

$$y = f(w, x)$$

- ▶ What is really $f()$? (*This is what we need to find.*)
- ▶ Who gives w ? (*Data gives.*)
- ▶ Who gives the form of $f()$? (eg. Quadratic, linear etc.?)

Regression and Time Series

Regression and Time Series

Regression

- ▶ Predicting a real number is regression

Regression and Time Series

Regression

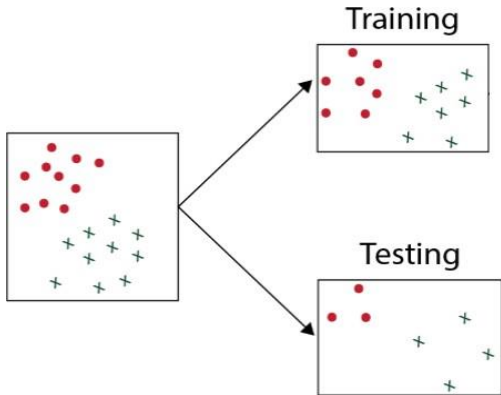
- ▶ Predicting a real number is regression

Time series Forecasting

- ▶ Predicting based on prior time tagged data

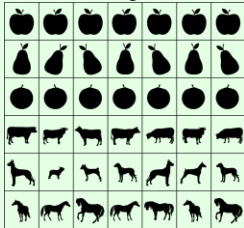
Training and Testing

Creating and Evaluating Models

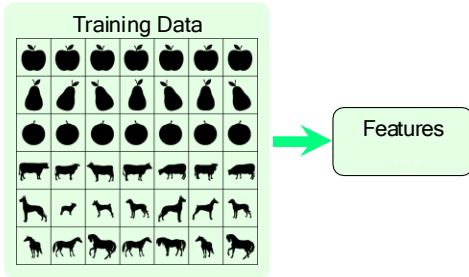


► Training

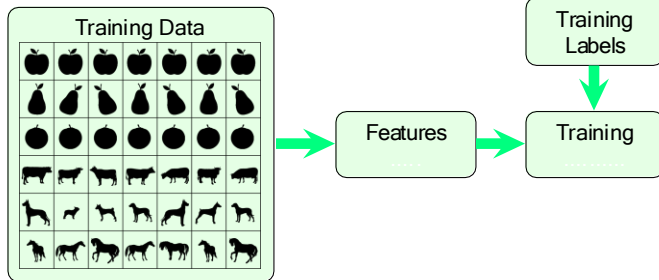
Training Data



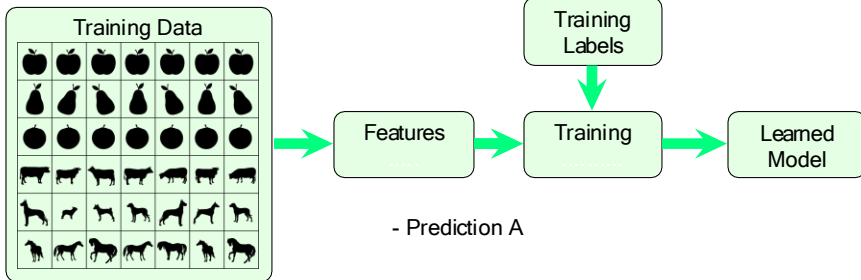
▶ Training



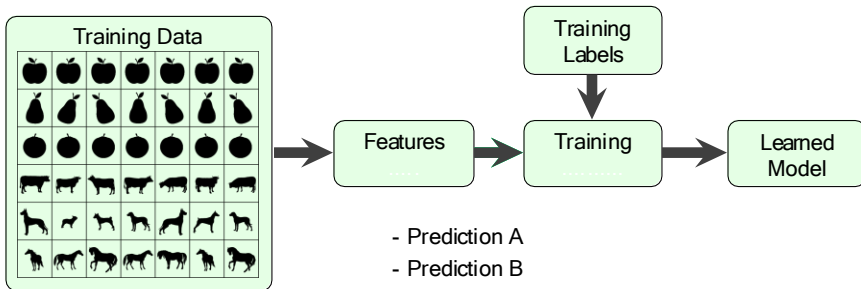
► Training



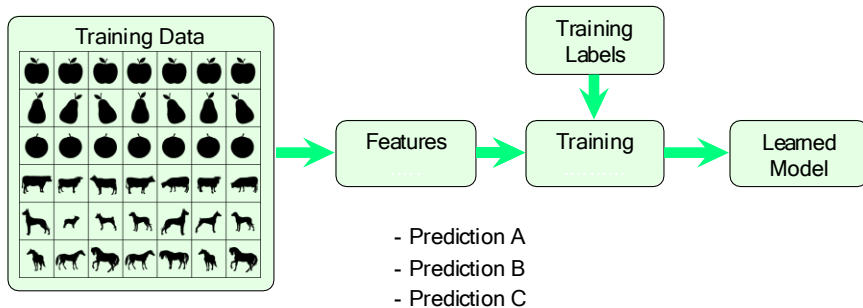
► Training



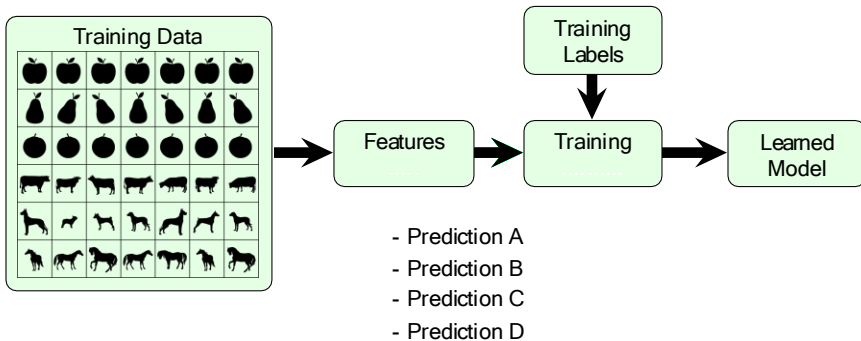
► Training



► Training



► Training



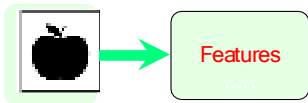
Steps ...

► Testing



Steps ...

► Testing



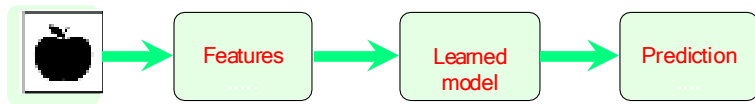
Steps ...

► Testing

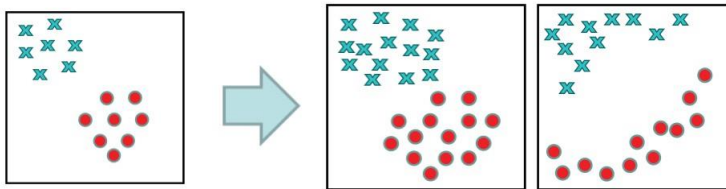


Steps ...

► Testing



- ▶ Training is the process of making the system able to learn
- ▶ **Assumptions:**
 - Training set and testing set come from the same distribution
 - Need to make some assumptions or bias



- ▶ Demo Train test split Penguin
 - ▶ Dataset: Penguin
 - ▶ Objective: To split the data into train and test sets using `train_test_split()` from `sklearn`

Two Prominent Learning Paradigms

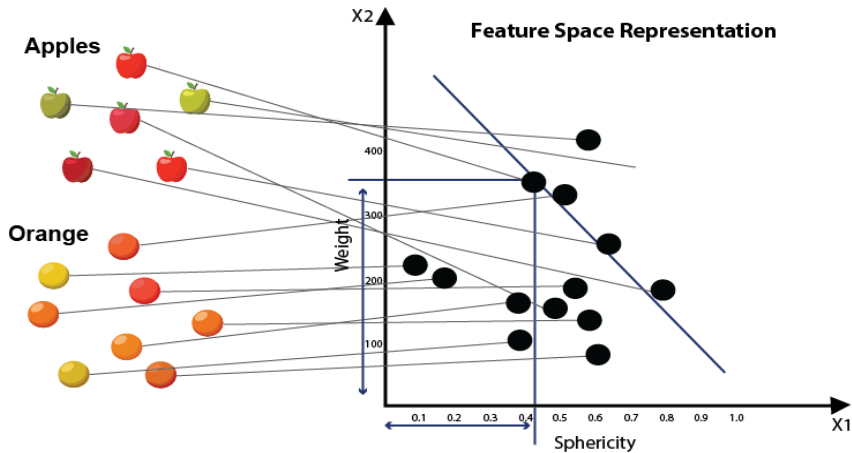
- ▶ **Supervised learning:** It is the machine learning task of inferring a function from labeled training data
- ▶ **Unsupervised Learning:** Learn patterns from unlabeled data. Often look for a structure

A 'toy' Classification Problem

- ▶ Apples vs Oranges
- ▶ We have measured weight, color, sphericity
- ▶ Some labeled data
- ▶ Given unlabeled data decide which fruit it is



Visualizing a Sample in 2D



Sample/Point and Representation

- ▶ A sample is easy to visualize in 2D $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$
- ▶ And sometime in 3D with some effort $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$
- ▶ And we often need much larger dimensionality in practice

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ x_d \end{bmatrix}$$

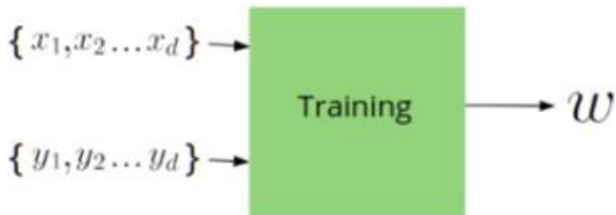
Examples of Learned Function or Model

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ x_d \end{bmatrix} \quad W = \begin{bmatrix} w_1 \\ w_2 \\ \cdot \\ w_d \end{bmatrix}$$

$$W \cdot X = w_1 \cdot x_1 + w_2 \cdot x_2 + \dots w_d \cdot x_d$$

$$f(w, x) = \text{sign}(w \cdot x)$$

In General ...



- ▶ Ground Truth
- ▶ Labels
- ▶ Predictions
- ▶ Training and Testing
- ▶ Supervised
- ▶ Unsupervised

- ▶ Features
- ▶ Input
- ▶ Output
- ▶ Feature Representation
- ▶ Samples
- ▶ Learning Model
- ▶ Classification